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Inferences

10 questions · ~15 min

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Q1

INFORMATION AND IDEAS

Parthenogenesis is a form of reproduction in which a female reproduces without mating. She produces an egg containing a single offspring whose DNA, or genetic material, comes entirely from its mother. Among birds, parthenogenesis has been found in a handful of species, including pigeons and turkeys. When scientists at the San Diego Zoo analyzed the DNA of the zoo's California condors (a species of vulture), they discovered that two individuals weren't genetically similar enough to any of the males in the condor enclosure to be their offspring. However, both had hatched from eggs laid by females in the enclosure. Thus, the scientists concluded that _____ blank

Which choice most logically completes the text?

- A. at least one of those individuals' mothers was born as a result of parthenogenesis.
- B. California condors may reproduce through parthenogenesis in zoos but not in the wild.
- C. the mothers of the two individuals probably reproduced through parthenogenesis.
- D. California condors reproduce through parthenogenesis only if females lack sufficient access to males.

In 2016 biological anthropologist Heather F. Smith and her team investigated the evolution of the appendix, an intestinal organ that is present in some mammals, including humans, but is generally thought to have no function. Studying 533 mammal species, the team found that the appendix has emerged independently across multiple lineages in separate instances and, significantly, hasn't disappeared after emerging in specific lineages. Moreover, the team determined that species with the organ tend to have higher concentrations of lymphoid tissue, which supports immune responses, in the cecum, the organ the appendix is attached to. Therefore, the team hypothesized that the appendix likely _____ blank

Which choice most logically completes the text?

- A. was once present in many nonmammal species but has since disappeared from those lineages.
- B. has been preserved in certain mammal species because it benefits their immune systems.
- C. will emerge in a greater number of mammal species because it may serve a necessary function in the immune system.
- D. produced higher concentrations of lymphoid tissue in mammals in the past than it does currently.

Astronomers investigated the Arabia Terra region of Mars because it appears to contain irregularly shaped craters that may have been caused by massive volcanic explosions. In their investigations of Arabia Terra, the researchers found remnants of ash deposits in an amount and thickness that would result from a massive volcanic eruption. However, erosion and past resurfacing events could have modified the surface of the planet. Therefore, _____ blank

Which choice most logically completes the text?

- A. the current makeup of the Arabia Terra region might not accurately reflect the volcanic activity of Mars's past.
- B. eruptions from Mars's volcanoes were likely not as massive as astronomers previously believed.
- C. ash was most likely expelled from multiple different volcanoes on Mars's surface.
- D. the craters found in the Arabia Terra region were necessarily created by events other than volcanic eruptions.

Euphorbia esula (leafy spurge) is a Eurasian plant that has become invasive in North America, where it displaces native vegetation and sickens cattle. *E. esula* can be controlled with chemical herbicides, but that approach can also kill harmless plants nearby. Recent research on introducing engineered DNA into plant species to inhibit their reproduction may offer a path toward exclusively targeting *E. esula*, consequently _____ blank

Which choice most logically completes the text?

- A. making individual *E. esula* plants more susceptible to existing chemical herbicides.
- B. enhancing the ecological benefits of *E. esula* in North America.
- C. enabling cattle to consume *E. esula* without becoming sick.
- D. reducing invasive *E. esula* numbers without harming other organisms.

The Indus River valley civilization flourished in South Asia from 3300 BCE to 1300 BCE. Many examples of the civilization's writing system exist, but researchers haven't yet deciphered it or identified which ancient language it represents. Nevertheless, archaeologists have found historical artifacts, such as clay figures and jewelry, that provide information about the civilization's customs and how its communities were organized. The archaeologists' findings therefore suggest that _____ blank

Which choice most logically completes the text?

- A. investigating an ancient civilization is easier without knowledge of the civilization's language.
- B. knowing an ancient civilization's language isn't necessary in order to learn details about the civilization.
- C. archaeological research should focus on finding additional artifacts rather than deciphering ancient languages.
- D. examining the civilization's historical artifacts has resolved the debate about this civilization's language.

Several artworks found among the ruins of the ancient Roman city of Pompeii depict a female figure fishing with a cupid nearby. Some scholars have asserted that the figure is the goddess Venus, since she is known to have been linked with cupids in Roman culture, but University of Leicester archaeologist Carla Brain suggests that cupids may have also been associated with fishing generally. The fact that a cupid is shown near the female figure, therefore, _____ blank

Which choice most logically completes the text?

- A. is not conclusive evidence that the figure is Venus.
- B. suggests that Venus was often depicted fishing.
- C. eliminates the possibility that the figure is Venus.
- D. would be difficult to account for if the figure is not Venus.

Although military veterans make up a small proportion of the total population of the United States, they occupy a significantly higher proportion of the jobs in the civilian government. One possible explanation for this disproportionate representation is that military service familiarizes people with certain organizational structures that are also reflected in the civilian government bureaucracy, and this familiarity thus _____ blank

Which choice most logically completes the text?

- A. makes civilian government jobs especially appealing to military veterans.
- B. alters the typical relationship between military service and subsequent career preferences.
- C. encourages nonveterans applying for civilian government jobs to consider military service instead.
- D. increases the number of civilian government jobs that require some amount of military experience to perform.

It's common for jazz musicians and fans to refer to certain songs as having "swing," indicating that the songs provoke a strong feeling, like the impulse to tap one's foot or dance. The exact acoustic properties that give a song swing, however, have long been thought to be undefinable. To investigate swing, a team led by physicist Corentin Nélis delayed the downbeats and synchronized the offbeats in jazz piano solos and asked jazz musicians to compare the intensity of swing in each modified piece with the intensity of swing in the original piece. They found that participants were more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing, suggesting that _____ blank

Which choice most logically completes the text?

- A. synchronized offbeats tend to give a song swing regardless of whether downbeats are delayed.
- B. the acoustic properties that give a song swing are not easy for jazz musicians to manipulate.
- C. jazz songs that feature the piano are more likely to have swing than are jazz songs that do not feature the piano.
- D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that _____ blank

Which choice most logically completes the text?

- A. dairy products were a more significant component of the Dutch diet of the period than lemons were.
- B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.
- C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.
- D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.

By running computer simulations of the development of our solar system, André Izidoro, Rajdeep Dasgupta, and colleagues concluded that the Sun may have been surrounded by three giant dust rings before the planets started to form. The researchers suggest that the materials in the innermost ring became the four planets closest to the Sun, the materials in the middle ring produced the rest of the planets, and the materials in the outermost ring created the asteroids and other small bodies in the region beyond Neptune. In one simulation, the researchers delayed the initial formation of the middle ring, causing oversized super-Earths to begin developing from the innermost ring. The researchers therefore hypothesize that _____ blank

Which choice most logically completes the text?

- A. the middle ring formed earlier in the solar system's development than the initial simulations suggested.
- B. the timing of the initial formation of the middle ring played an important role in determining the eventual size of Earth.
- C. if the formation of the outermost ring had occurred earlier in a simulation, all the planets would have become super-Earths.
- D. the innermost ring actually formed into all the planets in our solar system, not just the four closest to the Sun.